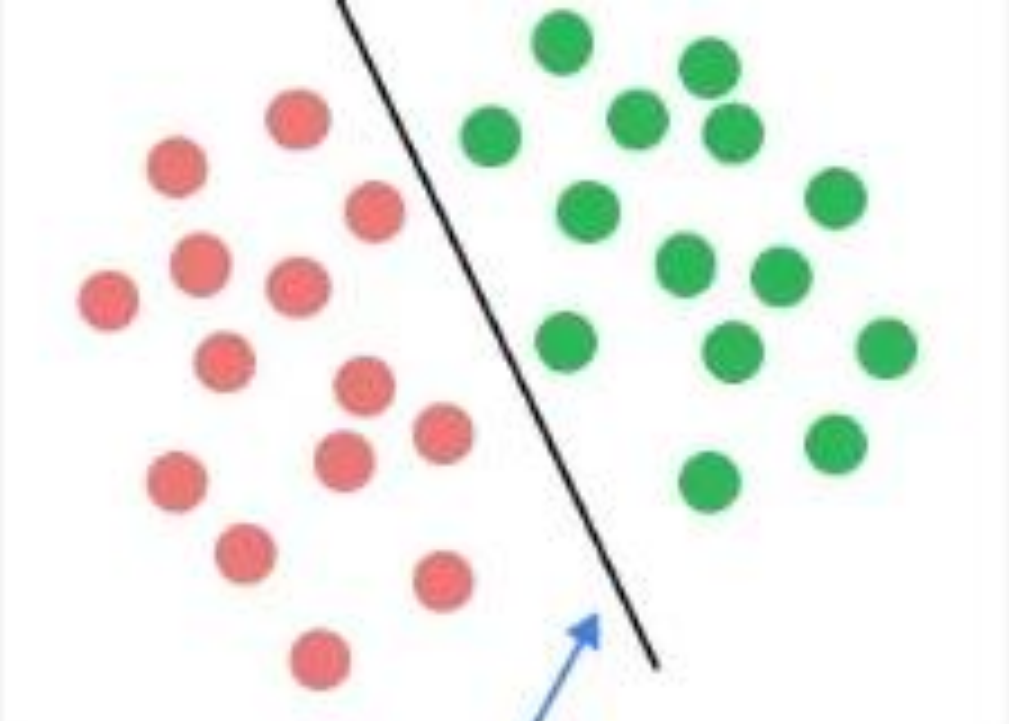
- Support Vector Machine is used for classification and regression.

- It works by finding decision planes that separate data into different classes.



Support Vector Machine (SVM)

- Support Vector Machine (SVM) is a widely-used supervised learning algorithm for classification and regression tasks in machine learning. Known for its robustness and ability to handle both linear and non-linear data, SVM has applications in fields ranging from healthcare to finance. Whether it's classifying images or detecting fraud, SVM offers a powerful solution by finding the optimal hyperplane to separate data into distinct classes
  A **Support Vector Machine (SVM)** is a supervised machine learning algorithm primarily used for classification tasks, although it can also be applied to regression problems. The core idea behind SVM is to find a **hyperplane** in an N-dimensional space (N representing the number of features) that best separates the data points into different classes. SVM is versatile enough to handle both **binary** and **multi-class classification** problems.

- In classification tasks, SVM seeks to identify the optimal hyperplane that maximizes the margin between different classes. This margin represents the distance between the hyperplane and the closest data points from each class, known as **support vectors**. The goal is to maximize the margin to ensure better generalization of the model.

- A real-world application of SVM can be found in **email spam detection**, where the algorithm classifies emails into "spam" or "not spam" based on various features extracted from the email content.

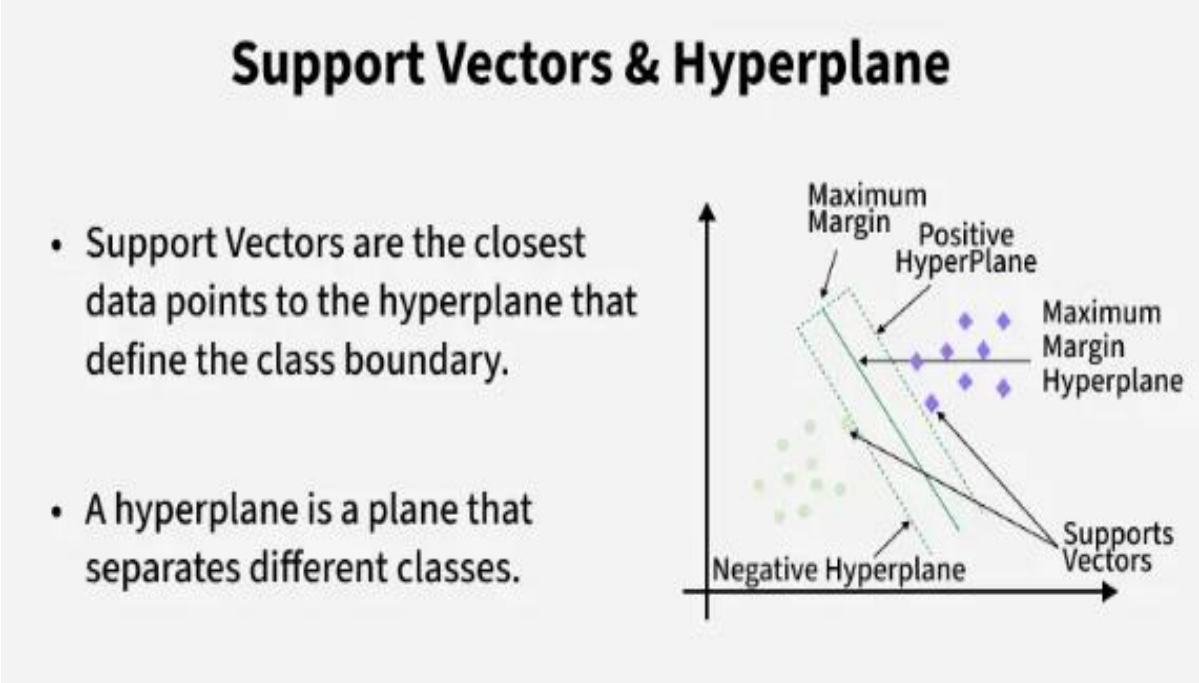
How Does the Support Vector Machine Algorithm Work?

- **Identify the Hyperplane**
  The first step is to identify a hyperplane that separates the classes. In two-dimensional space, this hyperplane is a straight line, but in higher-dimensional spaces, it can be a plane or a hyperplane. The goal is to select the hyperplane that maximizes the distance between the classes, ensuring that future data points are classified correctly.

- **Margins and Support Vectors**
  The margin refers to the distance between the hyperplane and the nearest data points from each class, which are called support vectors. Support vectors are critical data points that influence the position and orientation of the hyperplane. SVM seeks to maximize this margin, making the classification more robust.

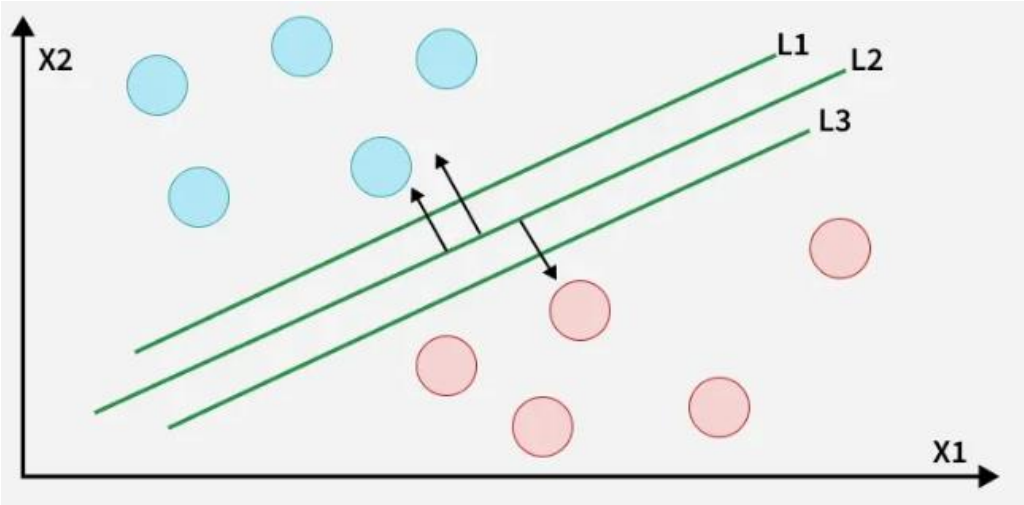
- **Maximizing the Margin**
  The optimal hyperplane is the one that maximizes the margin between the support vectors. A larger margin results in better generalization of the model, as it reduces the likelihood of misclassification when new data points are introduced.

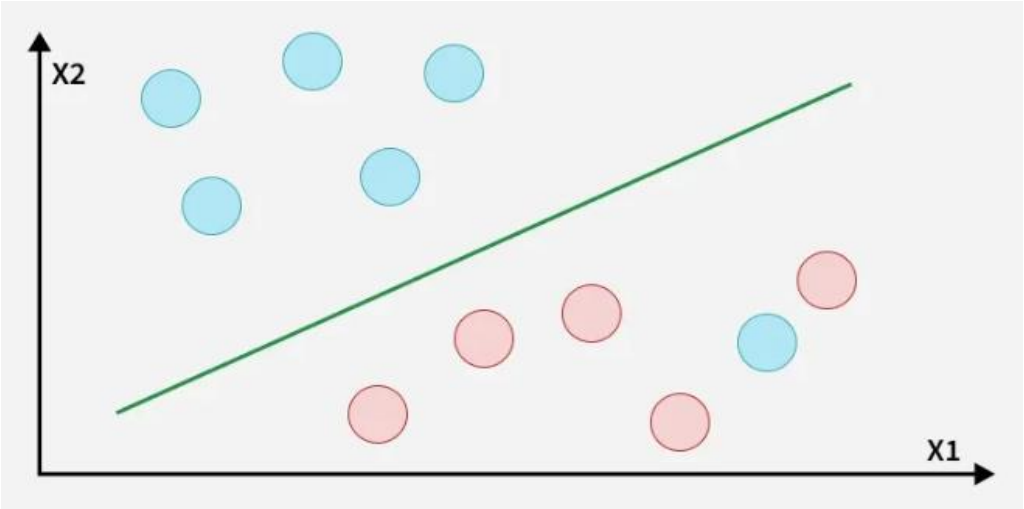
- **Decision Boundary**
  The decision boundary is defined by the hyperplane, and it is used to classify new data points. Any point that falls on one side of the hyperplane is assigned to one class, while points on the other side are assigned to the other class. For multi-class problems, SVM uses strategies like one-vs-one or one-vs-all to handle more than two classes.



How Does the Support Vector Machine Algorithm Work?

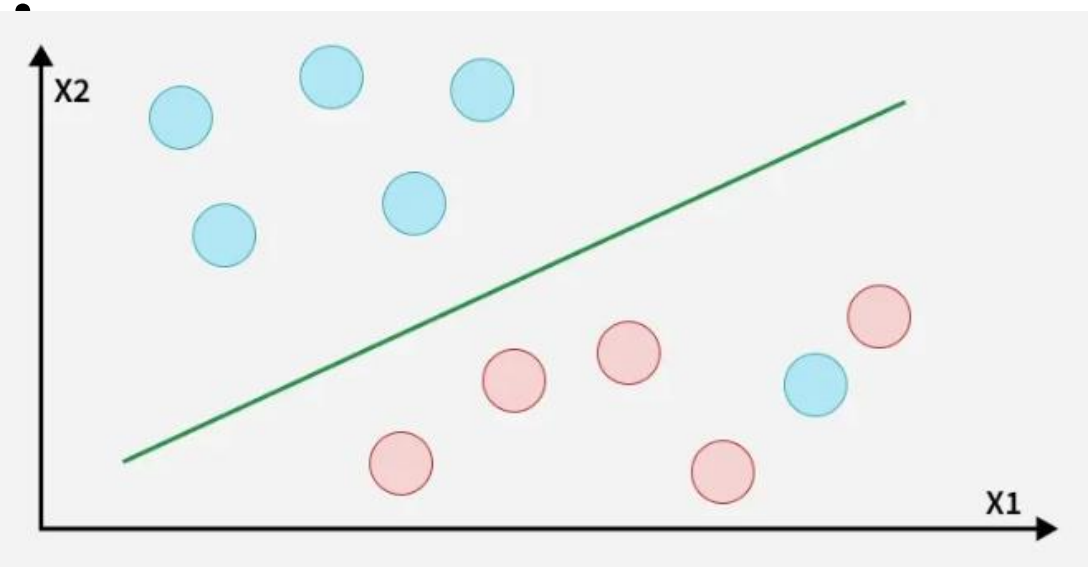
- The core idea of the SVM algorithm is to identify the hyperplane that separates two classes most effectively by maximizing the margin between them. This margin represents the distance from the hyperplane to the closest data points, known as support vectors, on either side.
- The optimal hyperplane, also referred to as the **hard margin**, is the one that maximizes the distance between the hyperplane and the closest data points from both classes. This maximization guarantees a clear separation between the classes. Based on the figure above, **L2** is selected as the hard margin. Now, let us consider a scenario as illustrated





How SVM classify data?

- The blue point located within the boundary of the red points is an outlier. The SVM algorithm is designed to handle such situations by ignoring the outlier and identifying the hyperplane that maximizes the margin. As a result, SVM is robust to outliers.
- A **soft margin** permits a limited number of misclassifications or margin violations in order to achieve better generalization. To accomplish this, SVM optimizes an objective function that balances maximizing the margin with minimizing the penalty for these violations.



$$\text{Objective Function} = \left(\frac{1}{\text{margin}}\right) + \lambda \sum \text{penalty}$$

The penalty applied for margin violations is commonly the **hinge loss**, which behaves as follows:

- If a data point is correctly classified and lies within the margin, no penalty is incurred (loss = 0).
- If a point is misclassified or violates the margin, the hinge loss increases in proportion to the extent of the violation.

Mathematical Computation of SVM

- Consider a binary classification task with two classes labeled **+1** and **−1**. The training dataset consists of input feature vectors **X** along with their corresponding class labels **Y**. A linear hyperplane that separates the classes can be expressed as:

$$w^T x + b = 0$$

Where:

- **w** is the normal vector to the hyperplane, indicating the direction perpendicular to it.
- **b** is the bias (or offset) term, which represents the distance of the hyperplane from the origin along the direction of **w**.

- The distance between a data point (x_i) and the decision boundary can be calculated as:

$$d_i = \frac{w^T x_i + b}{||w||}$$

Distance from a Data Point to the Hyperplane:

$$\hat{y} = \begin{cases} 1 & : w^T x + b \geq 0 \\ -1 & : w^T x + b < 0 \end{cases}$$

For a linearly separable dataset, the objective is to identify the hyperplane that maximizes the margin between the two classes while correctly classifying all data points. This objective results in the following optimization problem:

$$\underset{w,b}{\text{minimize}} \frac{1}{2} \|w\|^2$$

Subject to the constraint:

$$y_i(w^T x_i + b) \geq 1 \; for \; i = 1, 2, 3, \cdots, m$$

Where:

- (y_i) is the class label (+1 or −1) for each training instance.
- (x_i) is the feature vector of the (i)-th training instance.
- (m) denotes the total number of training instances.

The constraint (y_i (w^T x_i + b) > 1) ensures that each data point is correctly classified and lies outside the margin.

Soft Margin in Linear SVM

- When the data contains outliers or is not perfectly linearly separable, SVM permits some misclassifications by introducing **slack variables** (\zeta_i). The optimization problem is then modified as:

$$\underset{w,b}{\text{minimize}} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{m}\zeta_i$$

Subject to the constraints:

$$y_i(w^T x_i + b) \geq 1 - \zeta_i \quad \text{and} \quad \zeta_i \geq 0 \quad \text{for } i = 1, 2, \ldots, m$$

Where:

- C is a regularization parameter that controls the trade-off between margin maximization and penalty for misclassifications.
- ζ_i are slack variables that represent the degree of violation of the margin by each data point.